

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 12: Flight Hazards II: Thunderstorms & Microbursts

Syllabus Source: CAE Oxford (pp. 399-438) | Pass Target: >= 90% AviationExam

1. THE THUNDERSTORM (CB) LIFE CYCLE

A thunderstorm requires: (1) High atmospheric instability, (2) Abundant moisture, and (3) A lifting trigger (thermal, frontal, or orographic):

- 1. Cumulus Stage (Initial / Building): Purely UPDRAFTS (1,000 to 3,000 ft/min). Cloud grows vertically into a towering mass. No precipitation reaching ground.
- 2. Mature Stage (Peak Hazard): Marked by the ONSET OF PRECIPITATION at the surface! Updrafts (up to 6,000 ft/min) and downdrafts (up to 4,000 ft/min) coexist side-by-side. Severe turbulence, hail, lightning, microbursts. Duration: 20 to 30 minutes.
- 3. Dissipating Stage (Decay): DOWNDRAFTS PREDOMINATE throughout the cloud. The top freezes into a fibrous cirrus anvil (incus). Rain eases; cloud gradually dissipates.

2. THE MICROBURST PHENOMENON

A microburst is an extremely concentrated, violent downdraft descending from a convective cloud:

- Dimensions & Strength: Horizontal diameter < 4 km (2.2 NM). Downdraft speed can exceed 6,000 ft/min. Lifespan: 5 to 15 minutes.
- Aviation Flight Path Impact: An aircraft flying through a microburst encounters three rapid stages: (1) Severe HEADWIND increase (airspeed jumps, plane balloons above glide slope), (2) Severe DOWNBURST (vertical drop), and (3) Severe TAILWIND increase (airspeed plummets, catastrophic loss of lift and ground impact)!

3. AIRBORNE WEATHER RADAR (AWR) AVOIDANCE RULES

- Lateral Avoidance Margins: Avoid active CB cells by AT LEAST: 20 NAUTICAL MILES at or above FL 200; 10 NM below FL 200 (5 NM if temperature > 0°C).
- Vertical Avoidance: Avoid overflying an active CB by at least 1,000 ft for every 10 knots of wind speed at cloud top (or at least 5,000 ft above anvil)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Microburst Anatomy & Escape M

Microburst Encounter Profile on Final Approach: Step 1: Sudden strong HEADWIND -> Airspeed increases, aircraft balloons above glidepath. Pilot must NOT reduce thrust! Step 2: Center DOWNDRAFT -> Up to 6,000 ft/min vertical downdraft driving aircraft toward ground. Step 3: Sudden strong TAILWIND -> Airspeed drops 30-50 kt, massive loss of lift and stall hazard! Official Windshear Recovery Procedure: 1. Apply MAXIMUM TOGA THRUST immediately. 2. Disconnect autopilot and pitch smoothly toward Pitch Limit Indicator (PLI). 3. DO NOT CHANGE GEAR OR FLAP CONFIGURATION until safely established in positive climb!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What meteorological event formally marks the transition from the Cumulus stage to the MATURE STAGE of a thunderstorm?

- [A]: The first formation of an anvil
- [B]: The ONSET OF PRECIPITATION reaching the ground and the appearance of downdrafts
- [C]: Lightning seen in the distance
- [D]: Cloud top reaching FL 100

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Friction from falling rain drops drags air down, creating the first downdraft and marking the start of the mature stage.

Q2: When penetrating an active MICROBURST on final approach, what is the initial indication observed in the cockpit?

- [A]: Sudden severe loss of airspeed
- [B]: A SUDDEN INCREASE IN HEADWIND causing an abrupt increase in indicated airspeed and pitch-up above the glide path
- [C]: Stall warning
- [D]: Altimeter drops to zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The outflow ring produces a headwind first, causing airspeed to jump. If the pilot reduces thrust, the subsequent downdraft and tailwind will cause an unrecoverable crash.

Q3: What is the recommended minimum lateral separation from an active Cumulonimbus (CB) cell at FL 250?

- [A]: 5 NM
- [B]: AT LEAST 20 NAUTICAL MILES
- [C]: 2 NM
- [D]: 1 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At high flight levels (above FL 200), hail can be ejected miles outside the visible cloud column; 20 NM lateral clearance is standard airline SOP.